

p8106_hw1_sy3352

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```
library(glmnet)
library(caret)
library(tidymodels)
library(corrplot)
library(ggplot2)
library(plotmo)
```

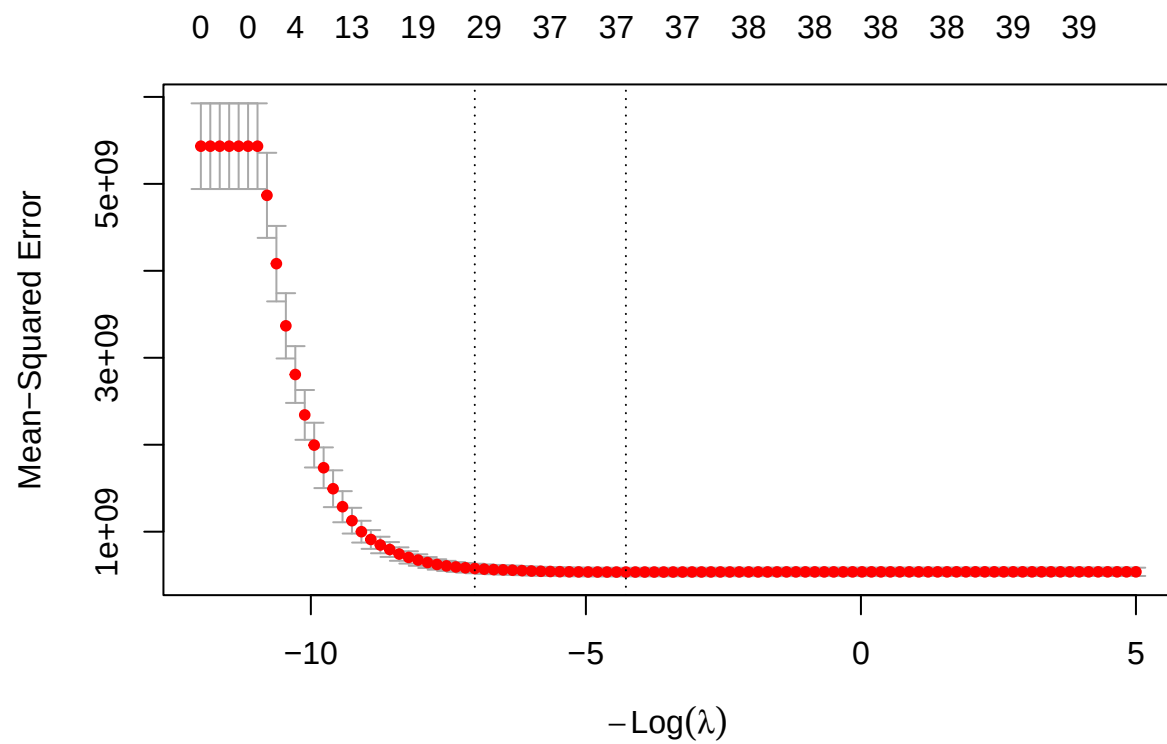
```
training_data <- read.csv("housing_training.csv") |> na.omit()
testing_data <- read.csv("housing_test.csv") |> na.omit()

x <- model.matrix(Sale_Price ~ ., training_data)[,-1]
y <- training_data[, "Sale_Price"]
```

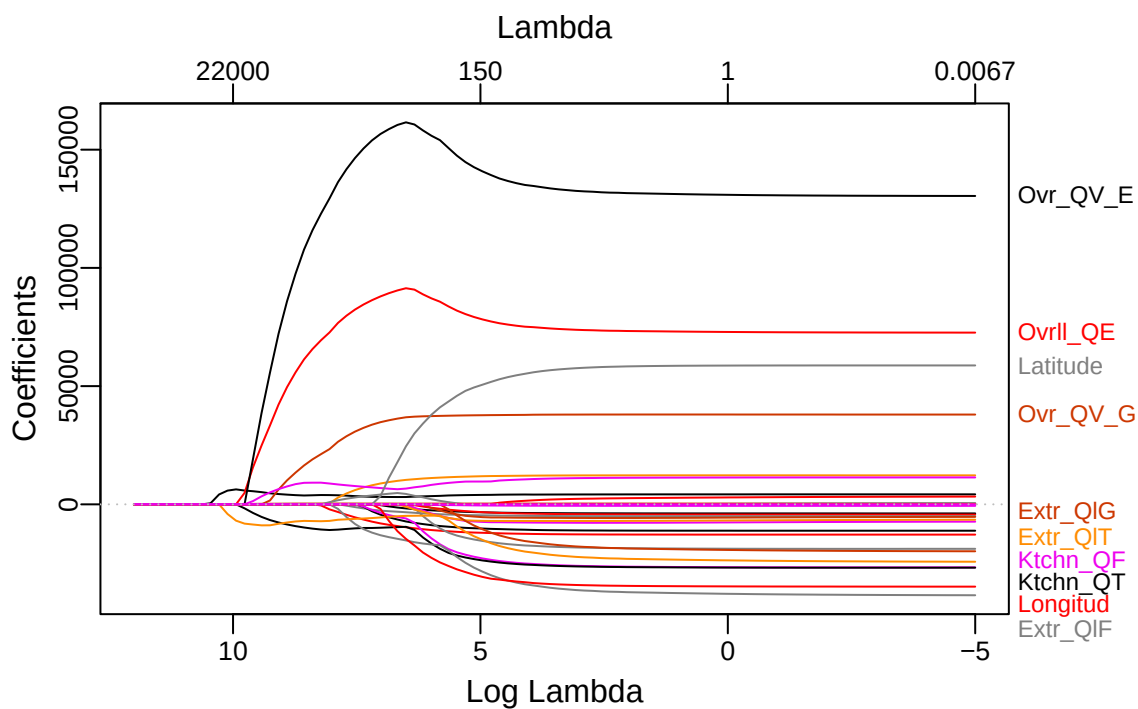
a)

```
set.seed(2026)
cv.lasso <- cv.glmnet(x, y,
                      alpha = 1,
                      lambda = exp(seq(12, -5, length = 100)))

plot(cv.lasso)
```



```
plot_glmnet(cv.lasso$glmnet.fit)
```



```
cv.lasso$lambda.1se
```

```
## [1] 1119.013
```

```
coef_1se <- predict(cv.lasso, s = "lambda.1se", type = "coefficients")
sum(coef_1se != 0) - 1
```

```
## [1] 29
```

```
lasso_pred <- predict(
  cv.lasso, newx = model.matrix(Sale_Price ~ ., testing_data)[,-1],
  s = "lambda.1se", type = "response")

lasso_test_RMSE = sqrt(mean((lasso_pred - testing_data$Sale_Price)^2))
lasso_test_RMSE
```

```
## [1] 20651.86
```

The selected tuning parameter is 1119.0126581. The test error is 2.0651861×10^4 . Using 1SE rule, the number of predictors included is 29.